

V&V11 — Statistical Assumptions

Normality, independence, equal variance

Shapiro-Wilk: applicable at N=20 — per-policy KPI distributions across replications.

→ For N=30 branch, walk-distance distributions across replications are checked.

Durbin-Watson: order-by-order autocorrelation of walk_m residuals.

→ Paired-by-order Wilcoxon (used in src/analyze.py) is rank-based and robust to autocorrelation in the differences.

Levene: equal-variance test across policies (used in Welch ANOVA fallback).

→ N=20 CRN replications (ASSUMPTIONS.md §24.2) — replication-based inference active.

